

- ① Solve (a) $\frac{dx}{dt} = t \cdot (x-2)$ with $x(0) = 5$.
(b) $\frac{dy}{dx} + 2xy = e^{-x^2}$ with $y(0) = 1$.
- ② A body originally at 60°C cools down to 40°C in 15 minutes when kept in air at a temperature of 25°C . What will be the temperature of the body at the end of 30 minutes.
- ③ The general solution of the differential equation $\frac{d^2y}{dx^2} + p \frac{dy}{dx} + q \cdot y = 0$ is given by $y(x) = a \cdot e^x + b \cdot e^{-3x}$. Obtain the values of the constants p and q .
- ④ Use Laplace transform to solve the initial value problem $\frac{d^2y}{dt^2} + 4y = 12t$ given that $y=0$ and $\frac{dy}{dt} = 9$ at $t=0$.
- ⑤ Given $F(s) = \frac{s+1}{s^2+2}$ is the Laplace transform of $f(t)$, find the Laplace transforms of (a) $\frac{f(t)}{e^t}$ (b) $3e^{2t} \cdot f(t)$ (c) $\frac{df}{dt}$.
- ⑥ Find the inverse Laplace transform of $F(s) = \frac{1}{s^2+25}$ using
(a) Partial fractions (b) convolution theorem
- ⑦ (a) Write down the general form of a second-order linear differential equation
(b) Characterize its general solution.
- ⑧ Solve the following linear system of differential equations using Laplace transform
- $$\begin{cases} x' - 2x - y' + 2y = -2t^2 + 7 \\ \frac{1}{2}x' + x - 3y' + y = t^2 + 6 \\ x(0) = 0 \quad y(0) = 6 \end{cases}$$
- ⑨

$$\begin{aligned} & x^n \cdot a^{n+1} \\ & -(x^2)^{n-2} \\ & x - x^2 \\ & e^{-(x-x^2)} \end{aligned}$$

END

Each question worth 5 Marks. Select any ~~five~~ Question only.